

Delta Drills

The Mathematical Model of the Mastery Engine

Concept graph $\times$ Bayesian Knowledge Tracing

Generated 2026-06-02 from backend/app/ (live code)

Every formula below is transcribed from `bkt_mastery.py`, `prioritization.py`, `concept_graph.py`, and `adaptive.py` — not idealized. Constants flagged “v0” are engineering defaults, not literature-derived.

1 The two objects and how they connect

Delta Drills runs on two coupled structures:

1. **The concept graph** $G = (\mathcal{A}, E)$ — a static, immutable-at-runtime directed acyclic graph. Nodes are *atoms* (drillable skills); edges are prerequisite relations, a subset of which are *encompassing*.
2. **The per-learner BKT state** — a time-stamped scalar belief per atom that the learner knows that atom. This is the only thing that changes as a student practices.

The graph is the fixed scaffolding; the BKT state is the moving belief that flows over it. The graph touches the belief in exactly two places: **forward (gating)** — prerequisite edges decide what a learner may see next (§4); and **backward (FIRE credit)** — encompassing edges route credit from a practiced atom to the simpler atoms it subsumes (§5.2). Everything else is a read of the BKT state (§6).

2 The concept graph G

2.1 Atoms (nodes)

$\mathcal{A} = \{a_1, \dots, a_N\}$, currently $N = 207$ (drillable graph). Each atom (**ConceptNode**) carries `id`, `title`, `topic`, `subtopic`, `kind` $\in \{\text{knowledge, skill, strategy}\}$, `tier`. Only `topic` enters the math (area scores, §6.4).

2.2 Prerequisite edges

An edge $e = (p \rightarrow d) \in E$ (**PrerequisiteEdge**) states atom p (prerequisite, simpler) precedes atom d (dependent, advanced).

Field	Symbol	Meaning
<code>prerequisite_id</code>	p	the simpler atom
<code>dependent_id</code>	d	the advanced atom
<code>is_encompassing</code>	$\iota_e \in \{0, 1\}$	does mastering d implicitly exercise p ?
<code>propagation_weight</code>	$w_e \in [0, 1]$	fraction of credit trickled $d \rightarrow p$
<code>is_hard_gate</code>	—	reserved; gating uses all edges
<code>weight, confidence, rationale</code>	—	metadata / provenance

Consistency invariant (enforced in `concept_graph.validate_graph`):

$$\iota_e = 0 \iff w_e = 0, \quad \iota_e = 1 \implies w_e > 0.$$

Thus encompassing $\subseteq$ prerequisite, and w_e is meaningful only on encompassing edges.

2.3 The two induced single-hop relations

Both are `lru_cached` (G is immutable at runtime).

Gating prerequisites (`_prereq_index`).

$$\text{Pre}(d) = \{p : (p \rightarrow d) \in E, p \notin \mathcal{A}_{\text{NG}}\},$$

where $\mathcal{A}_{\text{NG}}$ (`NON_GATING_ATOMS`) is a blocklist of infrastructure/dataset atoms (e.g. `cuda-availability-check`, `mnist-dataset`) that have no exercises and must never block learning.

Encompassed atoms (`_encompassing_index`).

$$\text{Enc}(d) = \{(p, w_e) : (p \rightarrow d) \in E, \iota_e = 1, w_e > 0\}.$$

Direction matters: $\text{Enc}(d)$ returns the *simpler* atoms p credited when the *advanced* atom d is practiced.

3 The learner state

For learner u at wall-clock time t , the entire algorithmic state is two maps over atoms (`UserPracticeState`):

$$L_a \in [0, 1] \text{ (atom_mastery)}, \quad \tau_a \text{ (atom_last_ts, ISO-8601)}.$$

L_a is the last-written BKT posterior $P(\text{learner knows } a)$; τ_a is when it was written. An untouched atom has no entry and is treated as the prior L_0 .

The legacy `SubtopicState` EWMA fields (`baseline`, `p`, `n`) still persist but are *no longer drivers* — since the BKT migration they are snapshots derived from the atom posteriors. The single source of truth is $\{(L_a, \tau_a)\}$.

3.1 Global constants (all v0 defaults, not literature-derived)

$L_0 = 0.10$	(P_INIT, prior $P(\text{known})$)
$T = 0.30$	(P_TRANSIT, learn rate / practice)
$g = 0.20$	(P_GUESS, $P(\text{correct} \mid \text{not known})$)
$s = 0.10$	(P_SLIP, $P(\text{wrong} \mid \text{known})$)
$\theta_m = 0.95$	(MASTERY_THRESHOLD)
$\theta_u = 0.85$	(UNLOCK_THRESHOLD)
$h = 14 \text{ days}$	(HALF_LIFE_DAYS)

4 The BKT core (per atom)

Three operators act on one atom’s belief: decay, observe, implicit-transit.

4.1 Forgetting / decay

For a stored belief L last written at τ , read at time t (**decay**):

$$\text{decay}(L, \tau, t) = L_0 + (L - L_0) \cdot 2^{-\Delta/h}, \quad \Delta = \max\left(0, \frac{t-\tau}{86400}\right) \text{ days}.$$

Missing $\tau \Rightarrow$ identity. As $\Delta \rightarrow \infty$, $L \rightarrow L_0$: stale knowledge reverts toward “unknown,” which resurfaces it for review (§6) without a separate staleness rule. The **decay-adjusted read** (`current_mastery`) is

$$\hat{L}_a(t) = \text{decay}(L_a, \tau_a, t), \quad \hat{L}_a(t) = L_0 \text{ if never practiced.}$$

Every downstream decision uses $\hat{L}$, never the raw stored L .

4.2 Evidence update (a graded attempt)

Corbett–Anderson (1994): Bayesian posterior under guess/slip, then learn-transit (**observe**). Posterior given the observation:

$$P(\text{known} \mid \text{correct}) = \frac{L(1-s)}{L(1-s) + (1-L)g}, \quad P(\text{known} \mid \text{wrong}) = \frac{Ls}{Ls + (1-L)(1-g)}.$$

With posterior L^* , the learn-transit step gives the new belief *assuming the attempt definitely exercised a*:

$$\text{observe}(L, x) = L^* + (1 - L^*)T.$$

4.3 Implicit-transit (one FIRE repetition)

An encompassed atom is never observed; it gets a learning step of size **gain** with no observation (`implicit_transit`):

$$\text{itransit}(L, \text{gain}) = L + (1 - L) \cdot \text{gain}, \quad \text{gain}, L \in [0, 1] \text{ (clamped).}$$

Same “move a fraction of the remaining distance to 1” shape as the transit term, driven by trickle-down credit rather than direct evidence.

5 The attempt pipeline and credit flow (backward use of the graph)

On a graded submission at time t , the handler (`feedback_router`) iterates the question’s **atom tags** — pairs (a, c) with tag confidence $c \in [0, 1]$ (how sure the question exercises a ; authored drill ≈ 1.0) — and calls `apply_attempt(a, correct, c)` for each.

5.1 The directly-practiced atom

Decay to now, apply evidence, then soft-apply by confidence (belief moves only a fraction c of the way to the full posterior):

$$\begin{aligned} L_a^- &= \text{decay}(L_a, \tau_a, t) && \text{(forgetting-adjusted prior)} \\ L_a^{\text{full}} &= \text{observe}(L_a^-, x) && \text{(if the tag were certain)} \\ L_a &\leftarrow L_a^- + (L_a^{\text{full}} - L_a^-) c && \text{(confidence-scaled write)} \\ \tau_a &\leftarrow t. \end{aligned}$$

$c = 1 \Rightarrow$ standard BKT update; $c = 0 \Rightarrow$ no-op.

5.2 FIRE credit to encompassed atoms (correct attempts only)

Iff the attempt was correct, set the implicit-rep magnitude gain $= T \cdot c$, and for every $(p, w_e) \in \text{Enc}(a)$ apply one implicit repetition against p ’s own decayed prior:

$$\begin{aligned} L_p^- &= \text{decay}(L_p, \tau_p, t), \\ L_p &\leftarrow \text{itransit}(L_p^-, \text{gain} \cdot w_e) = L_p^- + (1 - L_p^-) \text{clamp}(T c w_e), \\ \tau_p &\leftarrow t. \end{aligned}$$

Properties (by construction; asserted in `eg_validate.py`):

- Credit flows **advanced** $\rightarrow$ **simpler** only.
- Credit fires **only on correct** attempts.
- **Single-hop**: only atoms directly encompassed by a are credited; no transitive cascade in one attempt.
- **Bounded**: `itransit` keeps $L_p \leq 1$, and $w_e, c \leq 1 \Rightarrow$ implicit gain $\leq T$, so one encompassing rep moves p strictly less than a direct correct attempt.
- **Encompassing edges are the only channel sharing evidence between atoms**; with $E_{\text{enc}} = \emptyset$ this reduces to independent per-atom BKT.

5.3 What persists

`apply_attempt` mutates $\{L_a\}, \{\tau_a\}$ in place and returns the changed atoms (direct + trickled). That state is the entire algorithmic memory, serialized to the learner’s state file (Fly volume in prod).

6 Gating, selection & difficulty (forward use & reads)

6.1 Gating — forward use of the graph

Atom readiness (`atom_is_ready`): a is ready to be *learned* iff every gating prerequisite is cleared,

$$\text{Ready}_u(a, t) = \bigwedge_{p \in \text{Pre}(a)} [\hat{L}_p(t) \geq \theta_u].$$

Root atoms ($\text{Pre}(a) = \emptyset$) are always ready (cold-start entry points); the atom itself is not required (non-circular). The unified item gate (`item_is_unlocked`) for an item requiring atoms R is

$$\text{Unlocked}_u(R, t) = \bigwedge_{a \in R} [\hat{L}_a(t) \geq \theta_u],$$

with $R = \emptyset$ unlocked. This single rule replaced the old per-surface ad-hoc gates.

6.2 Subtopic mastery

Subtopic S exercises atoms $\mathcal{A}_S$ (from atom tags):

$$M_u(S, t) = \frac{1}{|\mathcal{A}_S|} \sum_{a \in \mathcal{A}_S} \hat{L}_a(t), \quad M_u(S, t) = L_0 \text{ if } \mathcal{A}_S = \emptyset.$$

6.3 Weakest-first selection

With effective weight ω_S (uniform $1/|S|$ unless customized):

$$\text{priority}_u(S) = \omega_S (1 - M_u(S, t)).$$

The next question is drawn from the unlocked, not-yet-exhausted subtopic of highest priority (alphabetical tiebreak). Un-practiced atoms sit at $L_0 = 0.10$, so fresh subtopics read weak and surface first; decay regresses mastered ones, resurfacing stale material through the *same* rule — Delta Drills’ “spaced repetition” is decay-driven resurfacing, not an SM-2/FSRS scheduler.

6.4 Target difficulty and area score

Difficulty scales affinely with mastery (`target_difficulty`, v0 map):

$$\text{diff}_u(S) = \text{clip}(20 + 80 M_u(S, t), 10, 100).$$

The learner-facing area score per topic T (`area_scores`, a pure readout) is the mean decayed posterior over practiced atoms,

$$\text{area}_u(T, t) = \frac{1}{|\mathcal{A}_T^{\text{seen}}|} \sum_{a \in \mathcal{A}_T^{\text{seen}}} \hat{L}_a(t).$$

7 End-to-end summary

The concept graph is a fixed DAG whose prerequisite edges gate *what you may practice* and whose encompassing sub-edges route *fractional credit backward* from advanced atoms to the simpler ones they subsume, while a decaying per-atom Bayesian Knowledge Tracing posterior is the only mutable state — updated by direct graded evidence and by single-hop FIRE trickle, and read back to pick the weakest unlocked atom to drill next.

8 Calibration caveat

Every constant in §3.1 and the affine maps in §6 are **v0 engineering defaults**, explicitly not literature-derived (see the `bkt_mastery.py` docstring and the 2026-05-24 source audit). Re-fit once real per-atom attempt data accumulates. The structure is the contract; the numbers are placeholders.